

Week 1

Exercise 1: Implementing the Singleton Pattern

```
package org.example;

//TIP To <b>Run</b> code, press <shortcut actionId="Run"/> or
// click the <icon src="AllIcons.Actions.Execute"/> icon in the gutter.
public class Main {
    public static void main(String[] args) {
        //TIP Press <shortcut actionId="ShowIntentionActions"/> with your
        caret at the highlighted text
        // to see how IntelliJ IDEA suggests fixing it.
        System.out.printf("Hello and welcome!");

        for (int i = 1; i <= 5; i++) {
            //TIP Press <shortcut actionId="Debug"/> to start debugging your
            code. We have set one <icon src="AllIcons.Debugger.Db_set_breakpoint"/>
            breakpoint
            // for you, but you can always add more by pressing <shortcut
            actionId="ToggleLineBreakpoint"/>.
            System.out.println("i = " + i);
        }
    }
}
```

Logger.java

```
package com.cognizant.singleton;

public class Logger {

    // Single instance of Logger
    private static Logger instance;

    // Private constructor
    private Logger() {
        System.out.println("Logger instance created.");
    }

    // Public method to access the instance
    public static Logger getInstance() {
        if (instance == null) {
            instance = new Logger();
        }
        return instance;
    }

    // Logging method
    public void log(String message) {
        System.out.println("Log: " + message);
    }
}
```

SingletonTest.java

```
package com.cognizant.singleton;

public class SingletonTest {

    public static void main(String[] args) {

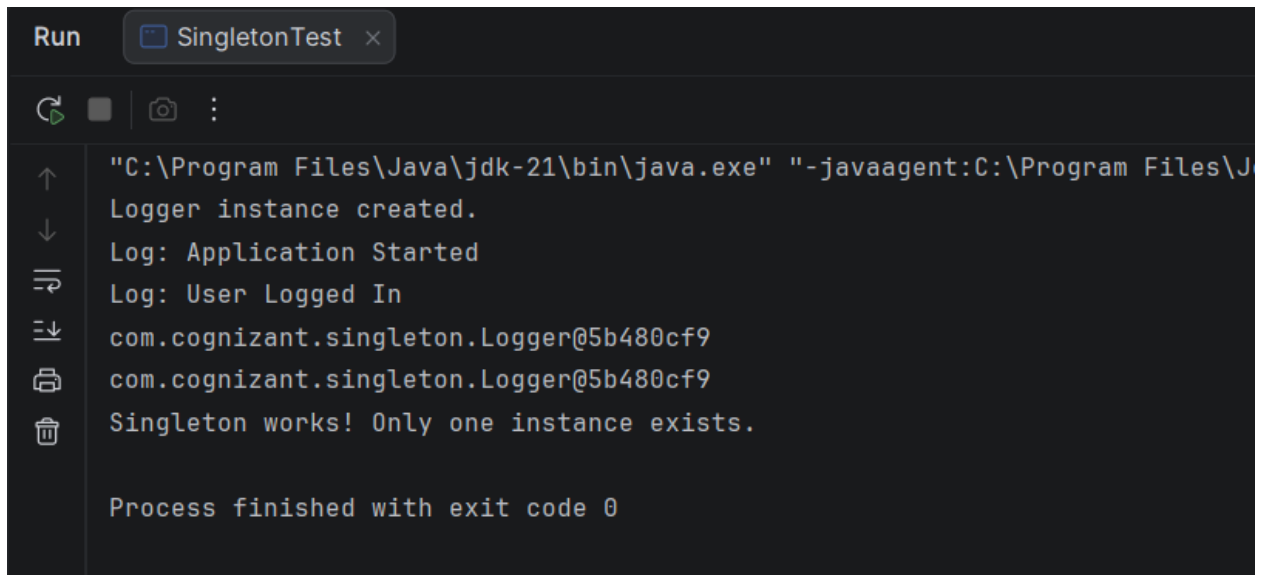
        Logger logger1 = Logger.getInstance();
        Logger logger2 = Logger.getInstance();

        logger1.log("Application Started");
        logger2.log("User Logged In");

        System.out.println(logger1);
        System.out.println(logger2);

        if (logger1 == logger2) {
            System.out.println("Singleton works! Only one instance exists.");
        } else {
            System.out.println("Singleton failed.");
        }
    }
}
```

Output



```
Run SingletonTest x
Logger instance created.
Log: Application Started
Log: User Logged In
com.cognizant.singleton.Logger@5b480cf9
com.cognizant.singleton.Logger@5b480cf9
Singleton works! Only one instance exists.

Process finished with exit code 0
```